Python Data Science

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Time required: 90 minutes

1. Save Python code in a Google Colab Notebook.

Tutorial 1: Plot Daily Weather in Scottsbluff

We are going to work with some daily weather data from NOAA (National Oceanic and Atmospheric Administration).

1. Go to: https://www.ncdc.noaa.gov/cdo-web/search
2. Select **Weather Observation Type/Dataset**
3. **Select Dataset**: Daily Summaries.
4. **Select Date Range**: A year ago to the current date. (You can select a much longer period if you wish)
5. **Search For:** ZIP Codes
6. **Enter a Search Term:** 69361 (or a different Zip Code if you wish.)
7. Click **Search.**
8. You will go a screen with a list of weather stations. **Scottsbluff, NE 69361 →** Click **Add To Cart.**
9. Click the **Cart (Free Data).**
10. **Select Cart Options → Custom GHCN-Daily CSV** (You can check and change the date range here if you wish.

11. Click **Continue.**

12. Custom Options: Daily Summaries → Select data types for customer output: **Air Temperature**. Click Continue.

13. You will get a page that says **Review Order**.

REQUESTED DATA REVIEW	
Dataset	Daily Summaries
Order Start Date	2024-01-01 00:00
Order End Date	2024-03-29 23:59
Output Format	Custom GHCN-Daily CSV
Data Types	TAVG, TMAX, TMIN
Custom Flag(s)	Station Name
Units	Standard
Stations/ Locations	SCOTTSBLUFF W B HEILIG FIELD AIRPORT, NE US (Station ID: GHCND:USW00024028)

14. Enter your email address to receive the link to your data.

15. **Submit Order**.

16. You should get a confirmation email that your order is being processed.

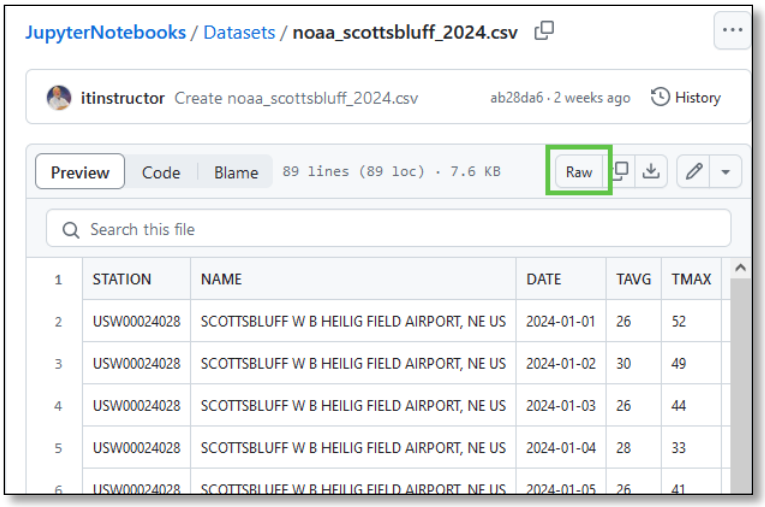
17. In a couple of minutes: You should get an email with a Download link.

18. Download the file as: **noaa_scottsbluff_2025.csv** in one of your GitHub repositories.

19. Commit the repository.

20. Got to the repository at www.github.com

21. Click on the csv file → Click the **raw** button.



22. Select and copy the url. It should look something like this.

https://raw.githubusercontent.com/itinstructor/JupyterNotebooks/main/Datasets/noaa_scottsbluff_2025.csv

Python Tutorial 1: Plot Daily Weather in Scottsbluff

1. In Google Colab → Create a Notebook named: **PythonNOAAScottsbluffDaily**
2. Enter the following code. Paste in your GitHub url for your csv file.

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the data from the CSV file "noaa_scottsbluff_2024.csv" into a DataFrame named "noaa".
noaa = pd.read_csv(
    "https://raw.githubusercontent.com/itinstructor/JupyterNotebooks/main/Datasets/noaa_scottsbluff_2024.csv"
)
```

```
# Display the first and last few rows of the DataFrame "noaa".
# This gives you an idea about the type of data
print(noaa.head())
print(noaa.tail())
```

If you end up with more than one weather station, you will want to filter down to one.

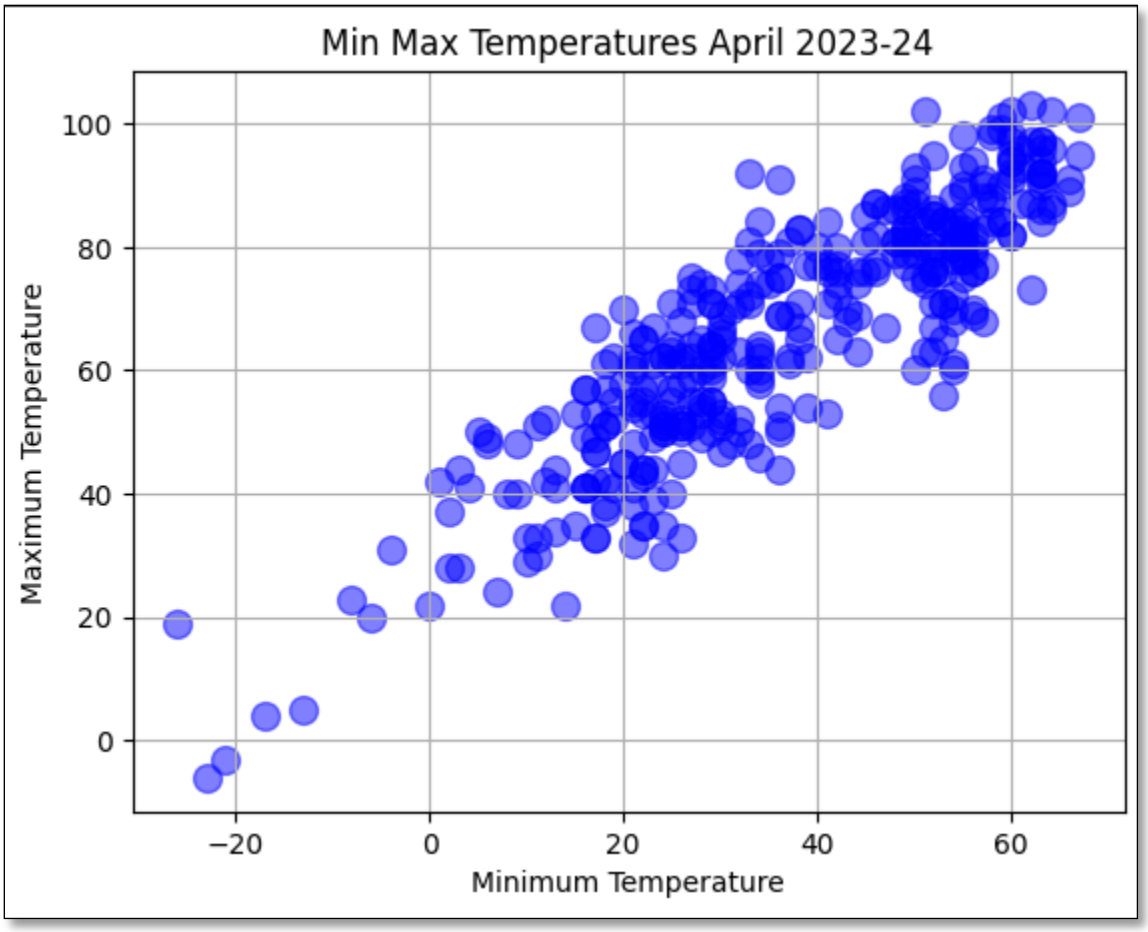
```
0  USW00094948  FAIRMONT MUNICIPAL AIRPORT, MN US  2024-01-01  32.0  19.0
1  USW00094948  FAIRMONT MUNICIPAL AIRPORT, MN US  2024-01-02  35.0  22.0
2  USW00094948  FAIRMONT MUNICIPAL AIRPORT, MN US  2024-01-03  29.0  20.0
3  USW00094948  FAIRMONT MUNICIPAL AIRPORT, MN US  2024-01-04  28.0  21.0
4  USW00094948  FAIRMONT MUNICIPAL AIRPORT, MN US  2024-01-05  35.0  21.0
         STATION                   NAME        DATE  TMAX  TMIN
334  US1MNMT0012  FAIRMONT 1.4 S, MN US  2024-04-19   NaN   NaN
335  US1MNMT0012  FAIRMONT 1.4 S, MN US  2024-04-20   NaN   NaN
336  US1MNMT0012  FAIRMONT 1.4 S, MN US  2024-04-21   NaN   NaN
337  US1MNMT0012  FAIRMONT 1.4 S, MN US  2024-04-22   NaN   NaN
338  US1MNMT0012  FAIRMONT 1.4 S, MN US  2024-04-23   NaN   NaN
```

```python
# Filter data to a single weather station
STATION = "USW00094948"
noaa = noaa.loc[noaa["STATION"].str.startswith(STATION)]
display(noaa.tail())
```

This section of code will extract and display a scatter chart.

```python
# Extract the 'TMAX' and 'TMIN' columns from the DataFrame "noaa"
# assign to variables
temp_max = noaa['TMAX']
temp_min = noaa['TMIN']

# Create a scatter plot with "temp_min" as the x-values,
# "temp_max" as the y-values, marker size of 100,
# filled markers in blue color with transparency set to 0.5.
plt.scatter(temp_min, temp_max, s=100, c="blue", alpha=0.5)

# Display the grid on the plot
plt.grid(True)

# Set the title of the plot to "Min Max Temperatures Jan-Mar 2024".
plt.title("Min Max Temperatures April 2023-24")

# Set the label for the x-axis as "Minimum Temperature".
plt.xlabel("Minimum Temperature")

# Set the label for the y-axis as "Maximum Temperature".
plt.ylabel("Maximum Temperature")

# Display the plot.
plt.show()
```

Example run:



Assignment 1: Weather Plots with Python

You may have to do some research to figure out how to do these. You can use other readings if you wish.

1. Create a line plot showing the TMIN and TMAX temperatures on the same plot.
2. Create a line plot showing the TAVG.
3. Create a bar plot showing the TAVG.

Assignment Submission

1. Insert a shared link to your Google Colab notebook that anyone can open.
2. Attach all to the assignment in Blackboard.